

Vibe Atlas — Unified System Design Document

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This document is a living specification. All sections are subject to revision as the system evolves through its four delivery phases. Changes must be reviewed by the relevant module owner and logged in the document version history.

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1. Executive Summary

Vibe Atlas is a content discovery and curation platform built around one core idea: quality determines placement. Rather than relying on engagement-farming signals or opaque editorial intuition, Vibe Atlas computes a transparent, multidimensional quality score - the Vibe Score - for every item and uses it as the authoritative signal for grid placement, search ranking, and partner content governance.

The system is composed of eight tightly-integrated modules, each sharing the Vibe Score as a common currency:

Module	Role
Grid Classification	Assigns content to visual slot tiers based on quality and freshness
Quality Metrics	Computes the Vibe Score from six weighted signals
Quality Heuristics	Applies rule-based guardrails before and after scoring

Module	Role
Promo Detection	Identifies and isolates promotional content
Height Emphasis	Gives high-quality mid-tier content extra visual presence
Search Strategy	Multi-signal ranking blending relevance, quality, and personalization
Connect Hub Integration	Governs partner content feeds, signal enrichment, and promo placements
UX Specification	Defines interaction design, component states, and accessibility standards

These modules share the Vibe Score as a common currency, reference each other's outputs at runtime, and are delivered across four phased milestones spanning approximately **40** weeks.

2. System Overview & Guiding Philosophy

2.1 The Problem Vibe Atlas Solves

Legacy content grids suffer from three core failure modes that Vibe Atlas is explicitly designed to address:

1. Engagement inflation. Items that game click-through mechanics surface regardless of actual quality, training users to distrust the feed and degrading the long-term signal value of engagement data.
2. Editorial bottlenecks. Manual curation does not scale; human overrides introduce inconsistency and latency that undermine the freshness and relevance users expect.
3. Promo opacity. Promotional content bleeds into organic positions with insufficient disclosure, eroding user trust and creating significant regulatory exposure under emerging sponsored-content frameworks.

Vibe Atlas addresses all three by creating a quality-first, algorithmically-governed system with transparent editorial controls and strict promotional segregation.

2.2 Core Design Principles

1. Quality as the universal language. Every decision - from grid slot assignment to search rank to promo filtering - is expressed in terms of the

Vibe Score or signals that contribute to it. There is no parallel scoring system or hidden editorial tier.

2. Algorithmic humility. The algorithm makes the first call; editorial humans make the final call. Overrides are first-class operations with audit trails, expiry windows, and cascade effects on dependent systems.
3. Transparency to users. Sponsored content is always labeled. Visual differentiation by slot size is real, not decorative. Personalization is opt-in and clearable by the user at any time.
4. Progressive trust for partners. New publishers receive provisional scores and limited placement until they establish a track record. Partner content is subject to the same quality floors as organic content, with no exceptions by default.
5. Privacy by architecture. Personalization signals are computed client-side where possible. No raw interaction logs leave the user's device. Users can inspect, correct, and delete their data at any time from Settings.

3. Architecture Overview

3.1 System Topology

The Vibe Atlas system is organized into five discrete layers. Each layer has clearly scoped responsibilities and communicates with adjacent layers through well-defined interfaces.

Layer	Name	Components & Responsibilities
Layer 1	Data Ingestion Layer	Content Publishers → Ingest API → Metadata Normalizer; Connect Hub Partners → CH API → Partner Adapter. Handles deduplication, field normalization, and initial routing.
Layer 2	Quality Computation Layer	Quality Heuristics (hard block, penalty, soft flag); Vibe Score Engine (6signal composite, cached per tier); Promo Detector (hard flags + ML confidence). All three operate in parallel postingest.
Layer 3	Classification & Ranking Layer	Grid Classifier (Tier 1-4 assignment, slot allocation); Height Emphasis (Tall / Extra Tall / Story, smart crop); Search Ranker (5-signal ranking + MMR diversity pass).

Layer	Name	Components & Responsibilities
Layer 4	Rendering Layer	Grid Layout Engine (12-col fluid, server-side layout); Search Results Assembler (promo interstitials); Connect Hub Card Injector (async).
Layer 5	Client / UI Layer	Grid Components (skeleton → real, height variants); Search UI

Layer	Name	Components & Responsibilities
		(filters, suggestions); Connect Hub Cards (contextual, async, partner badge); Editorial Overlay (pin UX).

3.2 Data Stores

Store	Technology	Purpose
Item Metadata	PostgreSQL	Canonical record for all content items
Vibe Score Cache	Redis	5 – min TTL (Hero/Featured), 15-min (Standard/Compact)
Search Full-Text Index	Elasticsearch	BM25 full-text retrieval
Search Semantic Index	Dense vector store	Embedding-based semantic retrieval
Quality Audit Log	Append-only log store	90-day retention; all scoring events
Promo Audit Log	Append-only log store	12-month retention; all promo classification events
Connect Hub Billing	Transactional DB	Impression events; daily reconciliation

3.3 Key Runtime Flows

Content Ingest Flow

1. Item arrives at Ingest API
2. Heuristic hard-block check (synchronous; rejection terminates pipeline)
3. Promo confidence computed (Stage 1 hard flags; Stage 2 probabilistic scoring)
4. Vibe Score computed (6 -signal weighted composite)

5. Height Emphasis assessed (eligibility check against grid context)
6. Grid tier assigned (Tier 1-4 classification)
7. Item written to metadata store (PostgreSQL)
8. Search index updated (Elasticsearch BM25 + vector embedding)
9. Grid layout cache invalidated (if Hero or Featured tier is affected)

Grid Render Flow

1. Request arrives at Rendering Layer
2. Server resolves layout (Hero $\times 1$, Featured ≤ 3 , Standard / Compact fill)
3. Height Emphasis assignments applied per resolved grid context
4. Skeleton payload sent to client (correct slot sizes pre-determined)
5. Real content hydrates into skeleton slots (fade-in: 200 ms)
6. Connect Hub contextual cards fetched asynchronously post-render
7. Cards injected into designated partner card positions

Search Query Flow

1. Query received at Search endpoint
2. Intent classified (Informational | Navigational | Transactional | Exploratory)
3. Entity extraction + ontology-based query expansion
4. Parallel retrieval: BM25 (top 200) + semantic (top 200) + structured filters (top 200)
5. Candidates merged and deduplicated $\rightarrow$ top 400 retained
6. Multi-signal ranking formula applied to all candidates
7. MMR diversity re-ranking pass ($\lambda = 0.7$)
8. Promo filtering: promos removed from organic slots
9. Results assembled with interstitials + Connect Hub cards
10. Response returned to client

4. Grid Classification

4.1 Purpose

Grid Classification translates the Vibe Score and associated signals into a concrete layout decision: which tier does this item occupy, and therefore what visual footprint does it receive? The classifier is the bridge between the abstract quality score and the tangible user experience of the content grid.

4.2 Classification Inputs

Dimension	Source
Vibe Score (0-100)	Quality Metrics engine
Content Type	Ingest metadata
Age (freshness)	publishedAt vs. now()
Engagement Density	Real-time interaction / impression ratio

4.3 Tier Definitions

Tier	Name	Vibe Score	Max Age	Grid Span
1	Hero	≥ 85	72 hours	Full-width
2	Featured	65-84	7 days	2-column
3	Standard	40-64	Unlimited	1-column
4	Compact	< 40 or Promo	Unlimited	Narrow / list

4.4 Classification Algorithm

```
function classifyItem(item): score = computeVibeScore(item) if
isPromo(item): return TIER_4 age = now() - item.publishedAt if score >=
85 and age <= 72h: return TIER_1 if score >= 65 and age <= 7d: return
TIER_2 if score >= 40: return TIER_3 return TIER_4
```

Video Bonus: Videos receive a +5 Vibe Score adjustment applied before threshold evaluation. This bonus is applied after all penalty heuristics but before classification.

4.5 Slot Allocation Rules

1. Hero exclusivity. Only one Hero slot per page. Ties broken first by Vibe Score, then by freshness (most recently published wins).

2. Featured cap. Maximum 25% of visible slots may be Featured-tier. Overflow items are demoted to Standard for layout purposes only.
3. Promo floor. Promotional items are hard-capped to Tier 4 regardless of any editorial override. No override mechanism can elevate a confirmed promo above Compact.
4. Height modifier. Tier 3 items with Height Emphasis may receive Featured-equivalent visual treatment but remain algorithmically Tier 3 for all scoring, reporting, and placement quota calculations.
5. Content type weighting. Videos receive a +5 Vibe Score bonus before thresholds are applied. This is the only content-type-specific adjustment in the classification algorithm.

4.6 Re-classification Triggers

Trigger	Affected Tiers	Latency
Scheduled refresh	Hero / Featured	Every 15 minutes
Scheduled refresh	Standard / Compact	Every 60 minutes
Engagement spike (> 3× baseline in 10 min)	All	Immediate
Editorial override	Any	Immediate; 24-hour expiry
Promo flag change	Any	Immediate
Item deleted / expired	Any	Within 15 minutes

4.7 Cold-Start Behavior

Newly published items with no engagement history receive provisional Tier 3 classification for the first 30 minutes. After the cold-start window, full Vibe Score computation runs on accumulated signals and full classification applies. Items from new publishers (fewer than 30 days on the platform) receive a provisional Source Authority Score of **50** until the first monthly editorial review.

4.8 Responsive Grid Breakpoints

Breakpoint	Columns	Hero span	Featured span	Standard span
xs (< 480px)	4	4	4	4
sm (480-767px)	6	6	6	3

Breakpoint	Columns	Hero span	Featured span	Standard span
md (7681023px)	8	8	4	4
lg (10241279px)	12	12	6	3
xl ($\geq$ 1280px)	12	12	6	3

5. Quality Metrics & The Vibe Score

5.1 The Vibe Score

The Vibe Score is the master quality signal for the entire system. It is a 0-100 floatingpoint value computed as a weighted sum of six normalized signals, each capturing a distinct dimension of content quality:

```
VibeScore = sum(weight_i * normalizedSignal_i) Clamped to [0, 100] after computation.
```

5.2 Signal Definitions & Weights

Signal 1: Engagement Rate - Weight: 0.28

```
engagementRate = (clicks + shares + saves + comments) / impressions
normalizedEngagement = min(engagementRate / targetEngagementRate, 1.0) * 100
```

Trailing window: 24 hours. Minimum impressions threshold: 50 (below this, the category average engagement rate is substituted to prevent cold-start inflation).

Signal 2: Completion Rate - Weight: 0.20

Applies to video and audio content only. Measures sessions reaching $\geq 80\%$ of total duration.

```
completionRate = completedSessions / startedSessions
normalizedCompletion = completionRate * 100
```

For non-video / non-audio content: signal is excluded and its weight is redistributed proportionally across remaining applicable signals.

Signal 3: Content Freshness — Weight: 0.18

$$\text{freshnessScore} = 100 * e^{(-\text{lambda} * \text{ageHours})}$$

Content Type	Lambda (decay constant)	Half-life (approx.)
Breaking news / Events	0.05	~14 hours
Standard editorial	0.02	~35 hours
Product listings	0.01	~69 hours
Evergreen / Reference	0.005	~138 hours

Signal 4: Source Authority — Weight: 0.15

Per-publisher trust score maintained by the editorial team. Range: 0-100. New publishers receive a provisional score of 50 . Score is updated monthly via editorial review and may be adjusted immediately in response to quality incidents.

Signal 5: Social Velocity - Weight: 0.10

```
velocity = (shareCount_t - shareCount_(t-2h)) / 2 normalizedVelocity
= min(velocity / categoryVelocityP95, 1.0) * 100
```

Signal 6: Visual Quality Score - Weight: 0.09

```
visualScore = (resolutionScore * 0.4) + (sharpnessScore * 0.3)
+ (compressionScore * 0.2) + (aspectRatioScore * 0.1)
```

Checks performed: resolution $\geq 800 \times 450px$; sharpness via Laplacian variance; JPEG compression artifact estimation; aspect ratio conformance to permitted values.

5.3 Weight Redistribution

When a signal is inapplicable (e.g., Completion Rate for a non-video item), its weight is redistributed proportionally across all remaining applicable signals:

```
redistributedWeight_i = weight_i + (excludedWeight * weight_i /
sum(weight_applicable))
```

5.4 Score Bands

Band	Range	Label	Grid Behavior
Exceptional	90-100	Top Vibe	Strongly Heroeligible
Strong	75-89	High Quality	Featured / Hero eligible
Good	60-74	Solid	Featured eligible
Moderate	45-59	Average	Standard
Weak	30-44	Below Average	Standard / Compact boundary
Poor	0-29	Low Quality	Compact; below 20 suppressed

5.5 Quality Floor & Promotional Ceiling

Quality floor: Items scoring below 20 are suppressed from the grid entirely. They are re-evaluated after 6 hours; items that remain below 20 after re-evaluation remain suppressed and are flagged for editorial review.

Promotional ceiling: Promotional items have an artificial score ceiling of 50 applied after Vibe Score computation but before classification. This prevents promos from achieving elevated grid placement through legitimate quality signals.

5.6 Cache & Recompute Policy

Tier	Cache TTL	Force-recompute Triggers
Hero / Featured	5 minutes	Engagement spike; editorial override
Standard / Compact	15 minutes	Engagement spike
All tiers	-	Editorial score pin (bypasses cache entirely; score treated as immutable for pin duration)

6. Quality Heuristics

6.1 Purpose & Relationship to Quality Metrics

Quality Metrics is the quantitative scoring layer - it answers "how good is this?" Quality Heuristics is the qualitative guardrail layer - it answers "should this even be here?" The two systems are complementary. An item can score well on the Vibe Score but still fail a hard-block heuristic and be permanently removed from the pipeline.

6.2 Heuristic Categories

Category	Enforcement	Effect
Hard Block	Automatic, no override	Permanently rejected from grid pipeline
Score Penalty	Automatic	Vibe Score reduced by defined penalty value (additive/stacking)
Soft Flag	Routes to editorial review	Item held pending manual review within defined SLA

6.3 Evaluation Order

1. Hard Block checks (HB-01 through HB-05) - synchronous, evaluated at ingest; rejection is terminal
2. Promo Detection (Stage 1 hard flags; Stage 2 probabilistic scoring)
3. Score Penalty heuristics (SP-01 through SP-07) - applied as deductions to computed Vibe Score
4. Soft Flag heuristics (SF-01 through SF-04) - asynchronous, routed to editorial review queue

6.4 Hard Block Heuristics

ID	Rule	Check / Method
HB-01	Broken / unreachable URL	HTTP 2xx within 5s; checked at ingest and every 6 hours post-publish
HB-02	Exact duplicate content	Content fingerprint uniqueness check against full catalog
HB-03	Prohibited content category	ML classifier + fact-check partner API call
HB-04	Malformed metadata	Required fields: title, canonicalUrl, publishedAt, contentType
HB-05	Publisher suspended	Publisher Registry state check at ingest and reclassification

6.5 Score Penalty Heuristics

All penalties are additive and stack. Total penalties may reduce Vibe Score below the quality floor, triggering suppression.

ID	Rule	Penalty
SP-01	Clickbait title pattern (NLP classifier, confidence $\geq$	-12

ID	Rule	Penalty
	0.75)	
SP-02	Thumbnail-content mismatch	-10
SP-03	Excessive interstitial ads (> 1 per 200 words)	-8
SP-04	Auto-play video with audio enabled	-15
SP-05	Outdated evergreen with time-sensitive claims (not updated > 18 months)	-6
SP-06	Thin content (article < 200 words)	-10
SP-07	Missing alt text on primary image	-3

6.6 Soft Flag Heuristics

ID	Rule	Review SLA	Possible Outcomes
SF-01	Near-duplicate content ($\geq$ 85% title similarity, same publisher, 24 h window)	2 business hours	Approve (original superseded); Approve (both serve); Reject
SF-02	Unverified breaking news (Source Authority Score < 70)	1 business hour	Approve; Reject; Escalate to factcheck partner
SF-03	Unusual engagement spike (> 10 $\times$ baseline in 30 minutes)	4 business hours	Approve (organic); Suppress (inauthentic); Escalate to Trust & Safety
SF-04	New and unvetted publisher (< 30 days on platform; first 5 items)	24 hours	Approve; Reject; Flag for extended review period

6.7 Heuristic Feedback Loop

Heuristic performance is reviewed on a monthly cadence. Editorial decisions on flagged items constitute ground-truth labels used to evaluate each heuristic's precision, recall, and false-positive rate. Results feed quarterly recalibration sessions where penalty values and classifier thresholds may be adjusted. All parameters are configurable in the `vibe-atlas.quality-heuristics` configuration namespace without service restart.

7. Promo Detection

7.1 Design Goals

1. Protect organic grid integrity from commercial content displacement
2. Ensure regulatory compliance on sponsored content disclosure
3. Maintain user trust through consistent and accurate visual differentiation
4. Enable auditable classification for billing, reporting, and dispute resolution

7.2 Detection Architecture - Stage 1: Explicit Hard Flags

Any one of the following signals unconditionally classifies an item as promotional (isPromo = true, label: EXPLICIT):

Signal	Detection Method
isSponsored: true in item metadata	Metadata field check
contentType: "advertisement"	Content type check
UTM parameter utm_medium=paid or utm_medium=cpc	URL query string parsing
Advertiser tag in publisher profile	Publisher Registry lookup
Affiliate link pattern in primary CTA URL	URL pattern matching against affiliate registry

7.3 Detection Architecture - Stage 2: Probabilistic Confidence Scoring

Signal	Weight
Superlative-heavy title ("Best", "Top 10", "Must-buy")	0.15
Brand name in title (brand registry match)	0.20
External tracking pixel detected in body	0.25
Price or discount mention in body text	0.15
CTA-heavy language ("Buy now", "Shop", "Get deal")	0.10
Publisher flagged as advertorial-heavy in registry	0.15

Decision logic:

Confidence $\geq 0.70 \rightarrow$ isPromo = true (ML_DETECTED) Confidence 0.50-0.69 -
> REVIEW_REQUIRED Confidence $< 0.50 \rightarrow$ isPromo = false

7.4 Promo Slot Rules

1. No Hero or Featured slots - hard prohibition with no override path available
2. Promo Compact slots - promotional items placed in the bottom 20% of the grid exclusively
3. Interstitial ratio - maximum 1 promo item per 6 organic items (1 : 6 ratio enforced)
4. Mandatory disclosure badge - always rendered; cannot be suppressed by publisher or editorial
5. Connect Hub Type 4 bypass - commercial promo placement partners operate under a separate placement protocol defined in Section 10.5

7.5 Disclosure Labels

Badge Text	Trigger Condition
Ad	Hard-flagged via advertiser metadata or content type field
Sponsored	Publisher-declared advertorial content
Partner Content	Connect Hub Type 4 commercial agreement placement
Promoted	Publisher self-declared promotional content

7.6 Review Queue & Dispute Resolution

Items in the REVIEW_REQUIRED band (confidence **0.50** — **0.69**) are treated as organic content pending editorial review, which must be completed within 4 business hours. Possible outcomes: CONFIRMED_PROMO, CONFIRMED_ORGANIC, or ESCALATE to compliance.

Publisher dispute process: Publisher submits dispute via Publisher Portal → logged with timestamp → editorial review within 2 business days → upheld disputes (confirmed organic) generate a negative feedback signal to the probabilistic confidence model, improving future precision.

7.7 Audit Trail

Every promo classification event is logged immutably with the following fields: itemId, timestamp, confidenceScore, confidenceLabel, triggeredSignals[], finalClassification. Logs are retained for 12 months and are available to billing, compliance, and editorial teams with role-based access controls.

8. Height Emphasis

8.1 Concept

Height Emphasis is the system's mechanism for expressing relative quality within a tier. Two Standard-tier items may have meaningfully different Vibe Scores; Height Emphasis gives the higher-quality item a taller visual footprint without promoting it algorithmically or inflating its tier assignment. It is a visual signal, not a ranking signal.

8.2 Height Tiers

Height Tier	Height	Row Span	Assignment Trigger
None (Standard)	240px	1	Default for all items
Tall	360px	2	Vibe Score ≥ 55 + VQS ≥ 70 + compatible aspect ratio
Extra Tall	480 px	2 (full)	Vibe Score ≥ 70 + VQS ≥ 70 + compatible aspect ratio
Story	720 px	3	Editorial override only (max 7-day expiry)

8.3 Eligibility & Assignment Algorithm

All of the following criteria must be satisfied for Tall or Extra Tall assignment:

53. Vibe Score threshold: ≥ 55 for Tall; ≥ 70 for Extra Tall

54. Visual Quality Score ≥ 70

55. Asset aspect ratio supports tall crop without loss of essential information

56. Item is not promotional

57. Grid density: fewer than 30% of items in current view are already heightemphasized

```
function assignHeightEmphasis(item, gridContext): if isPromo(item): return  
HEIGHT_NONE if item.editorialHeightOverride: return  
item.editorialHeightOverride if gridContext.currentTallItemRatio >= 0.30:
```

```
return HEIGHT_NONE vibe = item.vibeScore vqs = item.visualQualityScore  
if vibe >= 70 and vqs >= 70 and isCompatibleAspectRatio(item, "EXTRA_TALL"):  
return HEIGHT_EXTRA_TALL if vibe >= 55 and vqs >= 70 and
```

```
isCompatibleAspectRatio(item, "TALL"): return HEIGHT_TALL return  
HEIGHT_NONE
```

8.4 Smart Crop Logic

Smart Crop runs at ingest time and is cached - it introduces zero render-time latency. The process:

- 58. Face and subject detection via on-device ML inference
- 59. Saliency heatmap generated for the full image
- 60. Crop anchor set at the saliency region centroid
- 61. 10% safety padding applied on all sides to prevent edge clipping
- 62. Quality gate: if the resulting crop VQS falls below 60, the item is downgraded one height tier

8.5 Layout Interaction Rules

- Height emphasis never widens an item horizontally - all height-emphasized items remain single-column width
- Extra Tall item: 3 Standard-height items are stacked adjacently in the right column to fill the same vertical space
- Tall item: 2 Standard-height items are stacked adjacently in the right column
- Minimum 2-column gap enforced between height-emphasized items within the same row group (cascade prevention)
- Mobile: Extra Tall collapses to Standard on xs (< 480px); Tall is preserved on sm (480-767px)

8.6 Relationship to Grid Classification

Height Emphasis is visually expressive but algorithmically transparent. A Tier 3 item with Extra Tall height assignment still classifies as Tier 3 for all scoring, reporting, slot quota calculations, and analytics attribution. Tier 4 (Compact) items cannot receive any Height Emphasis assignment.

9. Search Strategy

9.1 Overview

The Vibe Atlas search system extends the quality-first philosophy into the query experience. It reuses the Vibe Score as a first-class ranking signal alongside

text relevance, freshness, and personalization. The same quality that earns a Hero slot in the grid also drives search ranking - ensuring consistency of quality signal across every surface in the product.

9.2 Query Pipeline

Stage 1 - Query Analysis

Intent classification (Informational | Navigational | Transactional | Exploratory); entity extraction; query expansion via Vibe Atlas ontology; spell correction; temporal signal detection (e.g., "today", "this week").

Stage 2 - Candidate Retrieval (Parallel)

Three parallel retrieval streams each return top 200 candidates: BM25 full-text index; dense vector semantic index; structured attribute filters. Results are merged and deduplicated → top 400 candidates retained for ranking.

Stage 3 - Multi-Signal Ranking

The full ranking formula is applied to all 400 candidates. Intent-adjusted weights are loaded based on the classified query intent from Stage 1.

Stage 4 - MMR Diversity Re-ranking

Maximal Marginal Relevance applied to the ranked candidate set to enforce publisher and category diversity constraints.

Stage 5 - Promo Filtering & Insertion

Promotional items removed from organic result slots. Promo items re-inserted at maximum 1:6 interstitial ratio.

Stage 6 - Results Assembly

Organic results assembled with interstitials, related search suggestions, and Connect Hub contextual cards (loaded asynchronously after initial render).

9.3 Ranking Signal Weights

```
rankScore = (alpha * textRelevance) + (beta *  
vibeScore_normalized) + (gamma * freshnessBoost) +
```

```
(delta * personalizationScore) + (epsilon * engagementVelocity)
```

Signal	Variable	Default Weight	Intent Adjustments
Text relevance (BM25 + semantic blend)	alpha	0.35	Breaking news: semantic component weight increases
Vibe Score (normalized 0-1)	beta	0.25	Exploratory: increases to 0.30
Freshness boost	gamma	0.18	Breaking: 0.35; Evergreen: 0.10; Transactional: 0.10
Personalization score	delta	0.14	Transactional: 0.22; Breaking news: 0.06
Engagement velocity	epsilon	0.08	—

9.4 Freshness Boost Function

$\text{freshnessBoost} = 1.0 + (\text{freshnessDecayScore} / 100) * \text{freshnessMaxBoost}$

Query Intent	freshnessMaxBoost
Breaking / time-sensitive	0.80
Standard	0.30
Evergreen	0.10

9.5 Personalization

Signal sources: Interaction history (30-day trailing window); explicit preferences (saved topics, followed publishers); session context (recency-weighted); location signals (where user-permitted).

Privacy guarantees: All signals are processed client-side where technically feasible. No raw interaction logs leave the user's device. Users may clear their personalization history at any time from the Settings panel. Personalization is opt-in; non-personalized results are the default state.

9.6 Query Expansion

Query expansion uses the Vibe Atlas ontology. Expansion terms are injected into the semantic query vector (not the BM25 query string) to avoid over-retrieval on literal token matches while benefiting from semantic breadth.

Illustrative examples:

- "coffee shops" expands to: café, espresso bar, brunch, co-working spaces
- "home workouts" expands to: bodyweight exercises, HIIT, yoga at home, no-gym fitness
- "weekend getaways" expands to: road trips, mini breaks, staycation, day trips

9.7 MMR Diversity Re-ranking

$\text{MMR_score}(d) = \text{lambda} * \text{relevance}(d, \text{query}) - (1 - \text{lambda}) * \text{max_similarity}(d, \text{already_selected})$ lambda = 0.7 (default)

Diversity constraints: Maximum 3 results per publisher in the top 10 positions; maximum 40% of results from any single content category in the top 10 positions.

9.8 Zero-Results Handling

When fewer than 3 results remain after all filtering steps:

63. Progressive filter relaxation applied in order: date filter → category filter → publisher filter (one relaxation at a time)

64. Semantic-only retrieval with reduced similarity threshold if relaxation is insufficient

65. "Did you mean [corrected query]?" surface if spell correction is available

66. "Explore instead" curated content block from entity-overlapping topics

A zero-results state with no actionable suggestion is never shown to users.

9.9 A/B Testing & Evaluation

Primary metric: nDCG@10 - target ≥ 0.72 (vs. baseline keyword-only search at 0.58).

Secondary metrics: Click-through rate (CTR); session depth; zero-result rate.

Traffic splits: **1% — 50%** per experiment depending on maturity and risk profile.

Config namespace: search.ranking.experiments

10. Connect Hub Integration

10.1 What is Connect Hub?

Connect Hub is the platform's partner ecosystem - the layer through which thirdparty publishers, data providers, and commercial partners interact with Vibe Atlas. It governs how external content and signals enter and move through the pipeline, with explicit rules for quality, placement, and disclosure at every level. Connect Hub does not create a privileged lane for partners; it creates a governed lane with defined accountability.

10.2 Partner Types

Type 1 - Content Partners

Editorial publishers contributing content via the Content Feed API. Subject to full Vibe Score computation, promo detection, and grid classification. No special placement they compete on quality. Source Authority Score ≥ 60 required to onboard.

Type 2 - Signal Enrichment Partners

Data providers contributing supplementary signals (off-platform engagement metrics, fact-check scores, audience insights). Signals blended into the Vibe Score at a configurable weight (default: 0.05 per partner, configurable up to 0.10 with Engineering approval).

Type 3 - Contextual Card Partners

Partners providing structured cards (prices, showtimes, ratings, business hours, weather, statistics) rendered inline in grid and search results. Cards are async-loaded, visually distinct from organic content, labeled "via [PartnerName]", and never displace organic item slots.

Type 4 - Promo Placement Partners

Commercial advertisers operating under Connect Hub agreements. Bypass the standard probabilistic promo detection pipeline but operate under a stricter placement protocol (see Section 10.5) with mandatory disclosure.

10.3 Content Feed API

Endpoint: POST /connect-hub/v1/feed/ingest

Authentication: Bearer token (per-partner key, scoped to partner type)

Required item fields: externalId, title, description, contentType, publishedAt, canonicalUrl, thumbnailUrl

Rate limits by partner tier:

Partner Tier	Items / Hour	Requests / Minute
Standard	5,000	60
Premium	25,000	300
Enterprise	Unlimited	1,200

Ingest quality gate - all must pass:

Requirement	Threshold
Source Authority Score	≥ 60
Thumbnail resolution	$\geq 800 \times 450\text{px}$
Title length	10-120 characters
Description length	≥ 50 characters
Content freshness	≤ 30 days old at ingest
Canonical URL reachability	HTTP 200 response within 5 seconds

10.4 Contextual Card API

Endpoint: GET /connect-hub/v1/cards?contextItemId={itemId}&partnerIds={list}

Each card response includes: partnerId, cardType (price / rating / showtime / hours / weather / stats), headline, body, cta (label + URL), displayUntil (ISO 8601 timestamp).

Cards expire after displayUntil and are suppressed silently - no error state is shown to users. Cards are fetched asynchronously after initial page render and injected via fade-in animation to avoid blocking the critical render path.

10.5 Type 4 Promo Placement Protocol

Firm rules for commercial promo placement partners:

- 67. Agreement-defined positions. Slot positions are specified in the partner agreement as [rowIndex, columnIndex] anchors - not determined algorithmically
- 68. Frequency cap. Maximum 1 promo per 8 organic items (stricter than standard 1:6 organic promo interstitial ratio)
- 69. Hero slot prohibition. Hero slot positions cannot be purchased. Maximum visual weight is Featured-equivalent

70. Mandatory disclosure. "Partner Content" label is always rendered; cannot be suppressed

71. Quality floor. Visual Quality Score 60 required to serve; items failing this requirement are withheld until compliant creative is submitted

72. Organic score isolation. Type 4 placements have no effect on organic Vibe Scores or grid classification of other items

Billing events are emitted per impression: partnerId, placementId, itemId, impressionTimestamp, viewabilityScore. Reconciled daily against partner billing systems.

10.6 Connect Hub in Search Results

Contextual partner cards appear in search results when all three conditions are met: (1) a card partner's data is relevant to top-ranked organic results, (2) query intent is classified as Transactional or Exploratory, and (3) card data freshness is within the displayUntil timestamp. Cards appear in a dedicated "From our partners" row, visually and positionally separate from both organic results and promo interstitials.

10.7 Monitoring & SLAs

Metric	SLA
Feed ingestion latency (p95)	≤ 2 seconds per item
Contextual card fetch latency (p95)	≤ 150 ms
Partner API availability	$\geq 99.5\%$ per calendar month
Billing event delivery guarantee	At-least-once; deduplicated within 24 hours

10.8 Circuit Breaker Behavior

1. Circuit opens after 5 consecutive failures or when p95 latency exceeds 500 ms
2. Falls back to cached last-known-good state (TTL: 30 minutes)
3. Stale cache expired: partner cards suppressed silently - no error state shown to users
4. Circuit closes after a successful health check probe (polled every 60 seconds)

10.9 Partner Onboarding Process

1. Application submitted via Connect Hub Partner Portal with business and technical contact information
2. Review by Editorial + Partnerships team within 5 business days
3. API credentials issued: key, rate tier assignment, and partner type designation
4. Integration testing completed in staging environment with test item set
5. Go-live approval issued by Partnerships team; production access granted

11. UX Specification

11.1 Design Principles

1. Content first. The grid exists to surface content; UI chrome should recede and never compete with items for visual attention.
2. Confidence through quality signals. Visual differentiation - slot size, height, position - must accurately communicate editorial quality. Decoration that does not reflect real quality differences is prohibited.
3. Transparency. Users must always understand why content appears and whether it is sponsored. No dark patterns in disclosure.
4. Progressive disclosure. Details surface on demand; the initial grid state is clean, scannable, and uncluttered.
5. Accessibility by default. WCAG 2.1 AA compliance is a baseline, nonnegotiable requirement - not a post-launch enhancement.

11.2 Grid Layout

Base Grid Specifications:

- Column system: 12-column fluid grid
- Gutter: 16px (sm-xl breakpoints); 12px (xs breakpoint)
- Row height unit: 240 px (standard row base)
- Grid padding: 24px horizontal (lg-xl); 16px (sm-md); 12px (xs)

11.3 Slot Component Variants

Variant	Thumbnail Ratio	Title Lines	Publisher	Engagement
Hero	21:9 or 16:9	3	Yes	Full metrics
Featured	16:9 or 4:3	2	Yes	Full metrics
Standard	4:3 or 1:1	2	Yes	Condensed
Compact	16:9 wide	1	Name only	None
Promo	16:9 or 4:3	2	Yes + disclosure label	None

11.4 Item States

Default State

Thumbnail at full opacity; title in primary text color; category tag visible; disclosure badge visible if promotional.

Hover State (Desktop)

Thumbnail scale $1.02 \times$ (200 ms ease); title color transitions to accent color; save icon fades in (150 ms); elevation: 04 px 16 px rgba (0, 0, 0, 0.12).

Focus State (Keyboard)

3px solid focus ring in brand accent color; focus ring offset 2px; no transform on focus (accessibility requirement - transforms on focus are disorienting for keyboard users).

Loading / Skeleton State

Animated gradient skeleton for thumbnail (left-to-right sweep); title (2-line block); publisher (1-line block). Skeleton dimensions match the resolved slot size to prevent layout shift.

Error State

Fallback category-specific illustration displayed in thumbnail area; title and publisher retained; page-level error boundary handles retry logic without full page reload.

11.5 Disclosure Badge Specification

Property	Value
Position	Top-left corner of thumbnail, inset
Background	#1A1A1A at 80% opacity (semitransparent dark)
Text color	White (#FFFFFF)
Font size	11px / 700 weight
Padding	3px 6px
Border radius	3px
Min contrast ratio	$\geq 4.5:1$ (WCAG AA for normal text)

Content Classification	Badge Text
Hard-flagged advertiser via metadata	Ad
Publisher-declared advertorial	Sponsored
Connect Hub Type 4 commercial placement	Partner Content
Editorial-promoted (pinned) organic content	No badge

11.6 Search UI

Search Bar

Fixed at top of page; full-width on mobile; centered at 640 px max-width on desktop; input debounced at 250 ms ; autocomplete suggestions triggered after 2 characters entered.

Autocomplete Panel

Maximum 8 suggestions; grouped into: Top Match, Recent Searches, Trending. Fully keyboard navigable (arrow keys, Enter to select, Esc to dismiss). Screen-reader accessible with ARIA live region announcements.

Filter Bar

| Filter | Options |

| :— | :— |

| Content Type | All, Video, Article, Image, Product, Event |

| Date | Any time, Past hour, Past 24 hours, Past week, Past month |

| Publisher | Searchable multi-select (inline search within filter panel) |

All active filters persist in URL query parameters for shareability and browser history support.

No-Results State

Show "Did you mean [spell-corrected query]?" if spell correction is available.
Show "Explore instead" curated content block with 6 items from related topics.
A completely empty state with no actionable suggestion is never presented to users.

11.7 Grid UX Flows

Initial Page Load

1. Server renders skeleton grid with correct slot sizes for expected tier distribution
2. First contentful paint: skeleton is visible; perceived layout is stable
3. Grid data hydrates: real items replace skeletons (fade-in: 200 ms per slot, staggered 20 ms)
4. Connect Hub contextual cards load async: injected with fade-in after all organic content is rendered

Infinite Scroll / Pagination

- Mobile: infinite scroll. Desktop: pagination (configurable per content vertical in CMS)
- Next page prefetch triggers when user scrolls to within 300 px of page bottom
- End of feed state: "You've reached the end" message + "Refresh feed" primary CTA

11.8 Height Emphasis UX Behavior

Tall Item Layout

Slot height: 360px. Thumbnail is smart-cropped vertically using cached crop anchor. Title expands to 3 -line display. Two standard-height items are stacked adjacently in the right column to fill equivalent vertical space.

Extra Tall Item Layout

Slot height: 480 px . Title renders at 3 lines with increased font size (18px vs. 15px standard). Three standard-height items are stacked adjacently in the right column.

Mobile Height Collapse

Extra Tall collapses to Standard on xs breakpoint ($< 480\text{px}$). Tall is preserved on sm breakpoint (480-767px). No layout shift on collapse - slot dimensions are resolved server-side based on detected breakpoint.

11.9 Accessibility Requirements

Requirement	Standard
Color contrast (text on background)	WCAG 2.1 AA: $\geq 4.5:1$ normal text; $\geq 3 : 1$ large text
Keyboard navigation	All interactive elements reachable via Tab; logical, predictable focus order
Screen reader support	All images have descriptive alt text; grid items have aria-label; ARIA live regions for dynamic content updates
Touch target size	$\geq 44 \times 44\text{px}$ for all interactive elements

Requirement	Standard
Motion	Respect prefers-reduced-motion: disable transforms and fade animations when set
Focus indicators	Always visible; never removed without a fully compliant custom replacement

11.10 Responsive Behavior Summary

Breakpoint	Grid Layout	Search Layout	Height Emphasis
xs ($< 480\text{px}$)	1-column, standard only	Full-width, no sidebar	Collapsed to standard
sm (480-767px)	2-column	Full-width, no sidebar	Tall preserved
md (768-1023px)	3-column	Full-width, cards below results	Full (Tall + Extra Tall)
lg (1024-1279px)	4-column	Sidebar layout for partner cards	Full (Tall + Extra Tall)
xl ($\geq 1280\text{px}$)	4-column + wider gutters	Sidebar layout for partner cards	Full (Tall + Extra Tall)

11.11 Interaction Patterns

Save / Bookmark

Optimistic save on tap/click (immediate UI state change before server confirmation). Haptic feedback on mobile. "Saved" toast notification at bottom of screen, 2 -second auto-dismiss. Saved state persists across sessions and syncs across devices for authenticated users.

Share

Native share sheet on mobile (Web Share API). Copy-link + platform-specific options popover on desktop. Share action logs an engagement event that contributes to Social Velocity signal.

Report / Flag

Accessible via item overflow menu (3-dot). Options: "Not interested", "Inaccurate", "Inappropriate", "Spam or scam". "Not interested" immediately suppresses the item and all items from that publisher for the current session and logs a negative preference signal.

12. Implementation Phases

12.1 Phase 0 - Foundation (Weeks 1-4)

Phase 0 - Foundation

Goals: Establish infrastructure, data pipeline, and configuration framework. No userfacing features ship in this phase.

Deliverables:

- Content ingestion pipeline with metadata normalization and validation
- Vibe Score computation service (initial signals: engagement rate, freshness, source authority)
- Grid layout engine (12-column fluid, all five breakpoints)
- Configuration management service (vibe-atlas.* namespace)
- Audit logging infrastructure with defined retention policies

- Local development environment and comprehensive seed data set

Accelerated Interactivity Track (Pulled Forward from Phase 2)

Three UX interactivity deliverables have been pulled forward from Phase 2 into Phase 0 to unblock frontend development and enable higher-fidelity testing earlier. These items run in parallel with foundation infrastructure work and do not create criticalpath dependencies.

Task	Originally Scheduled	New Schedule	Owner	Notes
Grid item hover & focus states	Phase 2 (Week 11)	Phase 0, Week 1	Frontend	Thumbnail scale 1.02x, title accent transition, save icon fade-in, WCAGcompliant focus ring
Skeleton loading animation	Phase 2 (Week 11)	Phase 0, Week 1	Frontend	Animated gradient sweep for thumbnail, title, and publisher slots; respects prefers-reducedmotion
Save / bookmark interaction	Phase 3 (Week 19)	Phase 0, Week 1	Frontend	Optimistic save, 2second toast confirmation, sessionpersisted state

All accelerated items are tracked on the Vibe Atlas Sprint Tracker and flagged with the s Accelerated tag.

Key Technical Decisions:

- Scoring cadence: 15 -minute refresh for Hero/Featured; 60-minute for Standard/Compact
- Cache layer: Redis with 5 -minute TTL for Hero/Featured scores

- Data stores: PostgreSQL (item metadata), Elasticsearch (search index)

Phase Exit Criteria:

- Pipeline processes 10,000 items/hour without performance degradation
- Vibe Score computed correctly for all content types in the seed set
- Grid renders correctly at all five defined breakpoints
- Config changes take effect within one refresh cycle with no service restart required
- Accelerated interactivity components (hover states, skeleton loader, save interaction) pass WCAG 2.1 AA audit and are reviewed by design team

12.2 Phase 1 - Core Grid & Quality (Weeks 5-10)

Phase 1 - Core Grid

&

Quality

Goals: Launch live Vibe Atlas grid with algorithmic classification, full quality metrics, and promo detection.

Deliverables:

- Grid Classification (all 4 tiers, slot allocation rules, cold-start handling)
- Full Quality Metrics signal set (all 6 weighted signals with weight redistribution)
- Promo Detection (explicit hard flags + probabilistic soft flags)
- Promo disclosure label rendering in all UI contexts
- Editorial override CMS integration (tier pin, height pin, score pin)
- Quality floor suppression (score < 20)
- Publisher authority score registry and management UI
- Cold-start handling (30-minute provisional classification)

Rollout Plan: Internal beta Week 8; External beta at 5% traffic Week 9; Full production launch Week 10.

Success Metrics:

- Hero slot accuracy: $\geq 95\%$ of Hero items meet Tier 1 eligibility criteria on manual audit
- Promo detection precision: $\geq 92\%$; recall: $\geq 88\%$
- Grid render p95 latency: ≤ 120 ms

Phase Exit Criteria: All deliverables accepted by QA; promo detection metrics met on holdout set; zero critical bugs in editorial override flow; rollback plan documented and validated in staging.

12.3 Phase 2 - Search & Height Emphasis (Weeks 11-18)

Phase 2 - Search & Height Emphasis

Goals: Layer search ranking and height emphasis onto the established quality foundation.

Deliverables:

- Multi-signal search ranking (all 5 signals, intent-adjusted weights)
- Semantic search index (dense vector embeddings, full catalog ingestion)
- Query analysis pipeline (intent classification, entity extraction, expansion, spell correction)
- Height Emphasis module (Tall, Extra Tall tiers, Smart Crop at ingest)
- Search analytics events pipeline
- Search A/B testing framework with traffic splitting controls
- MMR diversity re-ranking implementation
- Zero-results fallback flows (filter relaxation, semantic fallback, "Explore instead")
- Search UI components (results, filter bar, related queries, "did-you-mean")

Rollout Plan: Search runs in shadow mode (ranks candidates but does not serve) for 2 weeks before serving live traffic. Height Emphasis feature-flagged per content vertical. Full rollout Week 18.

Success Metrics:

- Search nDCG@10: ≥ 0.72 (baseline keyword search: 0.58)

- Search CTR improvement: $\geq 15\%$ vs. keyword baseline
- Zero-result rate: $\leq 4\%$
- Height Emphasis grid density cap maintained: $\leq 30\%$ tall items in any rendered view

12.4 Phase 3 - Personalization & Connect Hub (Weeks 19-28)

Phase 3 - Personalization

&

Connect Hub

Goals: Introduce user-level personalization, Connect Hub integration, and full analytics instrumentation.

Deliverables:

- Personalization signal computation (interaction history, explicit preferences, session context)
- Privacy-preserving personalization architecture (client-side signal processing)
- Connect Hub integration (content feeds, signal enrichment, contextual cards, promo bypass)
- Connect Hub editorial cards in grid and search results
- Personalization settings UI (history view, preference management, clear data)
- Full analytics instrumentation across all modules
- Weekly relevance evaluation report (automated pipeline)
- Publisher Portal (dispute submission, analytics dashboard, authority score visibility)

Rollout Plan: Personalization opt-in beta Weeks 19-22; Connect Hub integration Weeks 22-25; Full personalization rollout Week 26; Publisher Portal Week 28.

Success Metrics:

- Personalized session depth improvement: $\geq 20\%$ vs. non-personalized baseline
- Dispute resolution time: ≤ 2 business days
- User personalization opt-out rate: target $< 25\%$
- Privacy architecture audit: passed (engaged privacy counsel required in Phase 1)

12.5 Phase 4 - Optimization & Scale (Weeks 29-40)

Phase 4 - Optimization
&
Scale

Goals: Performance tuning, editorial tooling maturity, and scaling to full production load targets with $10\times$ headroom.

Deliverables:

- Search weight calibration from Phase 2/3 A/B experiment results
- Vibe Score signal tuning from production engagement data
- Grid layout performance optimization (sub- 80 ms p95 target)
- Editorial dashboard v2 (quality trends, override history, suppression management)
- Bulk editorial operations (pin/unpin, suppress batches, publisher-level actions)
- Load testing and capacity planning ($10\times$ traffic headroom validated)
- Technical documentation finalized and published (all modules)
- On-call runbooks published for all Vibe Atlas subsystems

Success Metrics:

- Grid render p95 latency: ≤ 80 ms
- Search p99 latency: ≤ 300 ms
- Vibe Score computation throughput: $\geq 50,000$ items per minute
- System availability: $\geq 99.9\%$ over any 30 -day window

12.6 Risk Register

Risk	Impact	Mitigation
Semantic index build time for full catalog	Phase 2 timeline delay	Begin embedding generation pipeline in Phase 0; run in background before Phase 2 needs it
Connect Hub partner API instability	Phase 3 quality degradation	Circuit breaker architecture; automatic fallback to organic content; stale cache TTL of 30 minutes
Promo detection false positive volume at launch	Publisher trust erosion	Robust dispute resolution flow; publisher communication playbook; model feedback loop from confirmed outcomes
Personalization privacy compliance delays	Legal / regulatory hold on Phase 3	Engage privacy counsel in Phase 1, not Phase 3; design privacy

Risk	Impact	Mitigation
		architecture before implementation begins
Cold-start quality degradation from new publisher content	Grid quality perception harm	Provisional scoring with conservative defaults; mandatory editorial review for new publishers' first 5 items
Accelerated UX work creates rework if design system changes in Phase 1	Minor visual inconsistency requiring CSS updates	Scope accelerated components to selfcontained CSS modules; design token-driven so updates propagate automatically

13. Cross-Cutting Concerns

13.1 Audit & Compliance

All module-level audit logs are written to append-only log stores with defined access controls and retention periods:

Log	Retention	Access
Quality (Vibe Score events)	90 days	Editorial, Engineering
Promo classification events	12 months	Billing, Compliance, Editorial
Editorial override events	24 months	Editorial, Compliance

Log	Retention	Access
Connect Hub billing events	36 months	Finance, Compliance

13.2 Configuration Management

All Vibe Atlas configuration lives under the `vibe-atlas.*` namespace in the central configuration service. Changes to the majority of parameters - including score weights, penalty values, and classifier thresholds - take effect on the next refresh cycle without any service restart.

Change approval requirements by parameter category:

- Score signal weights: Engineering approval
- Penalty values and soft-flag thresholds: Editorial approval
- Hard Block heuristic definitions: Engineering + Editorial joint approval
- Type 4 partner placement agreements (positions, frequency caps):

Partnerships + Legal joint approval

13.3 Observability

Core SLOs:

Metric	SLO
Grid render p95 latency	≤ 120 ms (Phase 1); ≤ 80 ms (Phase 4)
Search query p99 latency	≤ 300 ms
Vibe Score cache hit rate	$\geq 85\%$
Promo detection p95 latency (at ingest)	≤ 50 ms
Connect Hub card fetch p95	≤ 150 ms
System availability	$\geq 99.9\%$ per any 30 -day window

Alerting: All SLO breaches trigger PagerDuty alerts routed to the relevant on-call engineer. Runbooks for every alert type will be published by the end of Phase 4.

13.4 Data Retention Policy

Data Type	Retention	Deletion Method
Item metadata	While item is active + 90 days post-expiry	Soft delete at expiry; hard purge at 90-day mark
User interaction history	30 days rolling window	Automatic rolling deletion; user-initiated deletion available at any time

Data Type	Retention	Deletion Method
Personalization models	Rebuilt weekly from rolling interaction history	Previous model version archived for 14 days then purged
Search query logs	90 days	Automatic purge at 90day mark

14. Appendix - Algorithm Reference

This appendix consolidates all algorithmic formulas defined throughout this document into a single reference. Values shown are defaults; all configurable parameters reference their config namespace key.

14.1 Vibe Score Formula

```
VibeScore = (0.28 * normalizedEngagementRate) + (0.20 *
normalizedCompletionRate) # video/audio content only + (0.18 *
normalizedFreshness) + (0.15 * normalizedSourceAuthority)
+ (0.10 * normalizedSocialVelocity) + (0.09 *
normalizedVisualQuality) # All values normalized to [0, 100]. # Final score
clamped to [0, 100]. # Video/audio bonus: +5 applied after all penalties,
before classification. # Promo ceiling: max(score, 50) applied after
computation, before classification.
```

14.2 Classification Algorithm

```
function classifyItem(item): score = computeVibeScore(item) if
isPromo(item): return TIER_4 age = now() - item.publishedAt if score >=
85 and age <= 72h: return TIER_1 if score >= 65 and age <= 7d: return
TIER_2 if score >= 40: return TIER_3 return TIER_4 #
Videos: +5 to score before threshold evaluation. # Cold-start: TIER_3
provisional for first 30 minutes post-publish.
```

14.3 Height Emphasis Assignment

```
function assignHeightEmphasis(item, gridContext): if isPromo(item): return
HEIGHT_NONE if item.editorialHeightOverride: return
```

```

item.editorialHeightOverride if gridContext.currentTallItemRatio >= 0.30:
return HEIGHT_NONE if item.vibeScore >= 70 and item.vqs >= 70 and
isCompatibleAspectRatio(item, "EXTRA_TALL"): return HEIGHT_EXTRA_TALL
if item.vibeScore >= 55 and item.vqs >= 70 and
isCompatibleAspectRatio(item, "TALL"): return HEIGHT_TALL return
HEIGHT_NONE # Density cap: vibe-atlas.grid.heightDensityCap (default: 0.30)
# Smart Crop quality gate: if crop VQS < 60, downgrade one height tier.

```

14.4 Search Ranking Formula

```

rankScore = (0.35 * textRelevance) + (0.25 * vibeScore_normalized)
+ (0.18 * freshnessBoost) + (0.14 * personalizationScore)
+ (0.08 * engagementVelocity) freshnessBoost = 1.0 + (freshnessDecayScore /
100) * freshnessMaxBoost # Intent-adjusted weights override defaults (see
Section 9.3). # All weights sum to 1.0; intent adjustments maintain this
constraint.

```

14.5 MMR Re-ranking Formula

```

MMR_score(d) = lambda * relevance(d, query) - (1 - lambda) *
max_sim(d, already_selected_results) # lambda = 0.7 (default) - config key:
vibe-atlas.search.mmrLambda # Constraints: max 3 results per publisher in top
10; # max 40% per content category in top 10.

```

14.6 Promo Confidence Function

```

function computePromoConfidence(item): if hasHardFlag(item): return 1.0 #
EXPLICIT - no further computation needed score = sum(signal.weight for
signal in SOFT_SIGNALS if signal.detect(item)) return min(score, 1.0) #
PROBABILISTIC # Thresholds: # >= 0.70 -> isPromo = true (ML_DETECTED) #
0.50-0.69 -> REVIEW_REQUIRED (treated as organic pending review) # < 0.50
-> isPromo = false

```

14.7 Freshness Decay Function

```

freshnessScore = 100 * e^(-lambda * ageHours) # Lambda values by content
type: # Breaking news / Events: lambda = 0.05 (half-life ~14 hours) #
Standard editorial: lambda = 0.02 (half-life ~35 hours) # Product
listings: lambda = 0.01 (half-life ~69 hours) # Evergreen /
Reference: lambda = 0.005 (half-life ~138 hours)

```

14.8 Complete Parameter Reference Table

Parameter	Default Value	Config Key	Change Approval
-----------	---------------	------------	-----------------

:—	:—	:—	:—
promoConfidenceThreshold	0.70	vibeatlas.promo.confidenceThreshold	Editorial
reviewBandLower	0.50	vibeatlas.promo.reviewBandLower	Editorial
interstitialRatio	1:6	vibeatlas.promo.interstitialRatio	Editorial + Engineering
promoGridFloor	80%	vibeatlas.promo.gridFloor	Engineering
qualityFloorScore	20	vibeatlas.quality.floorScore	Editorial + Engineering

Parameter	Default Value	Config Key	Change Approval
		atlas.quality.floorScore	Engineering
promoScoreCeiling	50	vibeatlas.quality.promoScoreCeiling	Engineering
heroRefreshInterval	15 min	vibeatlas.grid.heroRefreshInterval	Engineering
standardRefreshInterval	60 min	vibeatlas.grid.standardRefreshInterval	Engineering
heightEmphasisDensityCap	0.30	vibeatlas.grid.heightDensityCap	Editorial
searchMMRLambda	0.70	vibeatlas.search.mmrlambda	Engineering
searchNDCGTarget	0.72	vibeatlas.search.ndcgTarget	Engineering
chCircuitBreakerThreshold	5 failures	vibeatlas.ch.circuitBreakerThreshold	Engineering
chCacheTTL	30 min	vibeatlas.ch.cacheTTL	Engineering
coldStartDuration	30 min	vibeatlas.grid.coldStartDuration	Engineering
newPublisherProvisionalAuthority	50	vibeatlas.quality.newPublisherAuthority	Editorial

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